Sentiment Analysis of Amazon Reviews using NLP and Machine Learning Models

Academic Portfolio Project
Completed at the University of Tasmania
Python, NLTK, Scikit-learn

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Abstract. Sentiment analysis of customer reviews provides valuable insights for e-commerce platforms. This project investigates sentiment classification of Amazon product reviews into positive, neutral, and negative categories. The dataset was pre-processed using NLP techniques including tokenization, stopword removal with negation handling, and lemmatization. Two feature extraction methods were compared: TF-IDF and CountVectorizer. The processed features were evaluated with Multinomial Naive Bayes and Linear Support Vector Machine classifiers. Experimental results show that SVM with TF-IDF achieved the best performance with an accuracy of 83.9%, while Naive Bayes performed better with CountVectorizer features (accuracy 82.0%). Across all models, the neutral class was challenging to classify due to imbalance. Overall, the findings confirm SVM as the most effective approach for sentiment classification of product reviews.
Keywords: Sentiment Analysis, Natural Language Processing (NLP), Text Classification.

1 Background

1.1 Motivation of the project

Sentiment analysis has become an essential area of natural language processing, enabling organisations to automatically interpret and respond to customer feedback. In the context of e-commerce, businesses rely heavily on online reviews to assess customer satisfaction and product performance. Manual inspection of thousands of reviews is infeasible; therefore, automated approaches provide significant value by offering fast and consistent insights. This project is motivated by the need to evaluate the effectiveness of traditional machine learning techniques in sentiment classification and to explore how different preprocessing and feature extraction strategies influence performance [1].

1.2 Objectives of the project

The objectives of this project are threefold. First, to apply natural language processing techniques to transform raw textual reviews into features suitable for machine learning.

Second, to design and implement at least two alternative approaches for sentiment classification by combining different feature extraction methods (TF-IDF and CountVectorizer) with two widely used classifiers (Naive Bayes and Support Vector Machine). Finally, the project aims to evaluate and compare these approaches using metrics such as accuracy and F1-score, and to identify the strengths, weaknesses, and practical implications of each method for three-class sentiment analysis.

1.3 Data Description

The dataset employed in this project, amazon-sampled.csv, was provided via the University of Tasmania's MyLO platform [2], consists of 4,000 records and six attributes: product name, brand name, price, rating, review text, and review votes. Several data quality issues were identified in the raw file. Specifically, missing values were found in the brand name (625), price (67), rating (5), review text (7), and review votes (117) attributes, along with 11 duplicate records. These inconsistencies were removed during preprocessing to ensure data reliability.

For the purpose of sentiment classification, only the review text and rating attributes were utilised, while other fields were excluded as they were not relevant to textual sentiment analysis. Ratings ranging from one to five stars were mapped into three sentiment categories: positive (4–5 stars), negative (1–2 stars), and neutral (3 stars). After cleaning, the dataset comprised 3,993 valid reviews, with the distribution heavily skewed towards positive sentiment (68.7%), followed by negative (23.8%) and neutral (7.5%). This imbalance, particularly the under-representation of the neutral class, presents a challenge for building robust classification models.

1.4 Research Questions

Based on the objectives above, the project is guided by the following research questions:

RQ1: Which combination of feature extraction and classification model (TF-IDF or CountVectorizer with Naive Bayes or SVM) provides the best performance for three-class sentiment analysis?

RQ2: How does the choice of feature extraction method (TF-IDF vs CountVectorizer) affect model performance with different classifiers?

RQ3: How does the class imbalance, particularly the under-representation of the neutral class, affect model performance?

2 Methodology

2.1 Data Pre-processing

Data cleaning: The raw dataset contained missing values in the brand name (625), price (67), rating (5), review text (7), and review votes (117) attributes, as well as 11 duplicate records. These were removed to ensure data quality.

Text preprocessing: Each review was lowercased, tokenised, and processed to remove stopwords, with the exception of negation terms (no, not, never). Words were lemmatised using WordNetLemmatizer. A negation-handling step was also implemented, in which tokens following a negation word were prefixed with "NOT_" until a punctuation mark was reached (e.g., "not good" → "NOT_good"), following recommendations in [3].

2.2 Feature Extraction

Two feature extraction approaches were adopted:

- **CountVectorizer** (Bag-of-Words): Represents each document by the frequency of words, disregarding grammar and word order, but preserving multiplicity. This simple yet effective representation has been widely used in information retrieval and text classification [4] [5].

- **TF-IDF** (Term Frequency–Inverse Document Frequency): Weighs words by both their frequency and their informativeness across the corpus, reducing the impact of common words. TF-IDF often provides better performance with SVM models [6].

In our implementation, both vectorisers were configured to accept pre-tokenised input (via a custom tokenizer and preprocessor) and to preserve negation handling. Default parameter settings were retained in order to maintain comparability across approaches.

2.3 Classification Models

Two classical machine learning classifiers were selected for comparison in this project: Multinomial Naive Bayes (NB) and Linear Support Vector Machine (SVM).

Multinomial Naive Bayes has historically been a strong baseline for text classification, particularly when features are represented as word counts. It is computationally efficient, easy to implement, and often performs surprisingly well despite its simplifying assumption of conditional independence among features [7]. This makes it a suitable candidate for sentiment analysis, where feature spaces are large but sparse.

Linear Support Vector Machines, on the other hand, are known to be highly effective for high-dimensional sparse data such as text [8]. By maximising the separating margin, SVMs provide strong generalisation performance and are robust to correlated and

redundant features. Numerous studies have shown that SVMs consistently outperform Naive Bayes and other traditional classifiers in sentiment classification tasks [9].

By comparing these two models, this study evaluates both a lightweight probabilistic baseline (NB) and a more powerful margin-based classifier (SVM), providing insights into the trade-offs between simplicity, efficiency, and predictive performance.

2.4 Experimental Setup

To evaluate the proposed approaches, the dataset was partitioned into training and testing subsets using an 80/20 split. Stratified sampling was applied to preserve the class distribution across the three sentiment categories, thereby ensuring that the minority neutral class was adequately represented in both subsets.

Performance was assessed using a range of evaluation metrics. In addition to overall accuracy, precision, recall, and F1-score were computed for each class. The macro-averaged F1-score was reported to account for class imbalance and to provide a more balanced assessment of model performance across positive, negative, and neutral categories [10].

All experiments were conducted using Python 3.12.4. The NLTK library was employed for natural language processing tasks such as tokenisation, stopword filtering, and lemmatisation, while Scikit-learn was used for vectorisation (CountVectorizer, TF-IDF) and classification (Naive Bayes, SVM).

To ensure reproducibility, all scripts and configuration files are included in the project submission. The dataset link is also provided in the appendix, enabling complete re-implementation and verification of the reported results.

2.5 Approaches Compared

Four approaches were tested:

- TF-IDF + Naive Bayes
- TF-IDF + SVM
- CountVectorizer + Naive Bayes
- CountVectorizer + SVM

This design allowed us to evaluate both the impact of feature representation and classifier choice on sentiment classification performance.
To ensure reproducibility, the full Python source code and dataset have been submitted alongside this report. The code is fully documented and structured, enabling re-implementation of the models and regeneration of all reported results. This provides transparency and guarantees that the experimental findings can be independently verified.

3 Evaluation

The performance of the four approaches—TF-IDF combined with Naive Bayes, TF-IDF combined with SVM, CountVectorizer combined with Naive Bayes, and CountVectorizer combined with SVM—was evaluated on the Amazon product reviews dataset. The target variable was sentiment, categorised into positive, neutral, and negative classes.

3.1 Performance Metrics

The models were assessed using standard evaluation metrics commonly applied in text classification:

- **Accuracy**: the overall proportion of correctly classified instances across all classes.
- **Precision**: the proportion of correctly predicted samples among all samples predicted as a given class.
- **Recall**: the proportion of correctly predicted samples among all actual samples of a given class.
- **F1-score**: the harmonic mean of precision and recall, providing a balanced measure between the two.

In order to account for the class imbalance present in the dataset, particularly the under-represented neutral class, both **macro-averaged** and **weighted-averaged** values of precision, recall, and F1-score were reported [Sokolova & Lapalme, 2009]. Macro averages assign equal importance to each class, while weighted averages adjust the contribution of each class according to its frequency in the dataset.

3.2 Results

Before presenting the classification performance, it is important to examine the distribution of sentiment classes in the dataset. As shown in Fig. 1, positive reviews dominate the dataset (68.7%), followed by negative reviews (23.8%), while the neutral class is severely under-represented at only 7.5%. This imbalance is expected to have a substantial impact on classification performance, particularly for the neutral class, as the models receive significantly fewer training examples from this category.

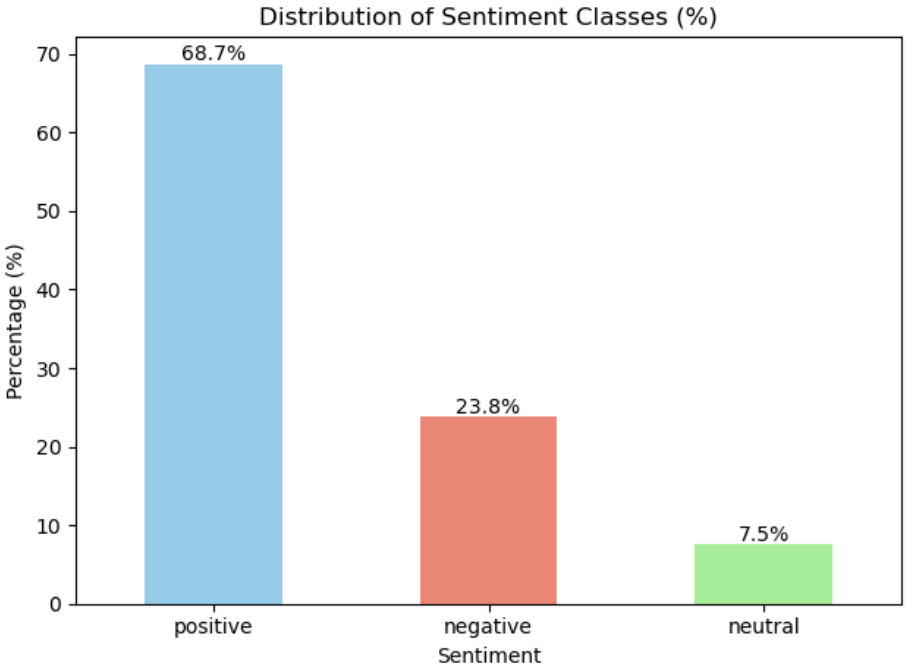

Fig. 1. Distribution of sentiment classes in the dataset.

Table 1 summarises the accuracy and macro-F1 scores for the four approaches. Figures 1–3 present the data distribution, model accuracy comparison, and confusion matrix of the best-performing model.

Table 1. Performance comparison of the four approaches.

Feature Representation	Classifier	Accuracy	Macro-F1
TF-IDF	Naive Bayes	0.73	0.38
TF-IDF	Linear SVM	0.84	0.60
CountVectorizer	Naive Bayes	0.82	0.54
CountVectorizer	Linear SVM	0.82	0.59

The results show that SVM classifiers outperformed Naive Bayes, and TF-IDF combined with SVM produced the best results with 84% accuracy and a macro-F1 of 0.60. Importantly, SVM also demonstrated stable performance across both feature representations, with CountVectorizer + SVM achieving 82% accuracy and a macro-F1 of 0.59, indicating that the model is robust to different vectorisation methods. In contrast, Naive Bayes was more sensitive to the choice of features: while it performed

poorly with TF-IDF (73% accuracy, 0.38 macro-F1), it achieved considerably stronger results with CountVectorizer (82% accuracy, 0.54 macro-F1).
As illustrated in Fig. 2, SVM-based models consistently achieved higher performance than Naive Bayes, regardless of the vectorisation method, while TF-IDF combined with Naive Bayes yielded the weakest performance.

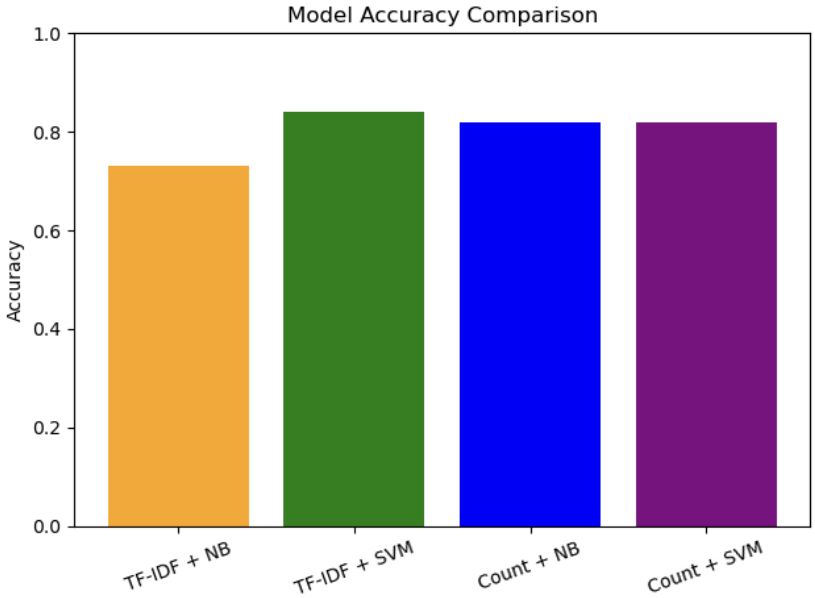

Fig. 2. Accuracy comparison of the four approaches.

Finally, Fig. 3 presents the confusion matrix of the TF-IDF + SVM model. Positive reviews were predicted with very high accuracy (500/549), and negative reviews were classified reasonably well (146/190). However, neutral reviews were almost always misclassified as positive or negative, with only 6 correct predictions out of 60. This highlights the effect of class imbalance: despite strong overall accuracy, the minority class was not effectively captured.

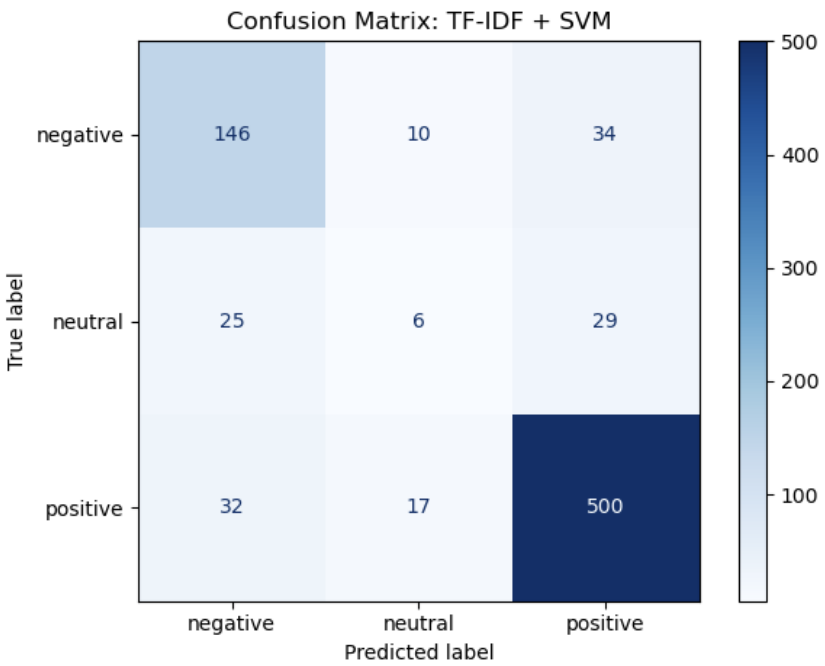

Fig. 3. Confusion matrix for the TF-IDF + SVM model (best overall).

3.3 Discussion

The results indicate that Linear SVM consistently outperformed Naive Bayes, particularly when combined with TF-IDF features, achieving the highest accuracy (84%) and macro-F1 score (0.60). This finding is consistent with prior research showing that SVMs are well-suited for high-dimensional sparse data such as text [8][9]. By contrast, Naive Bayes performed poorly with TF-IDF (73% accuracy), but achieved competitive performance with CountVectorizer (82% accuracy), supporting previous findings that frequency-based features align better with probabilistic models [7].

Despite these results, the neutral class remained problematic across all approaches. As shown in Fig. 3, only 6 out of 60 neutral reviews were correctly predicted, with most being misclassified as positive or negative. This is a direct consequence of class imbalance, as the neutral class accounted for less than 8% of the dataset. Consequently, the macro-F1 score was substantially lower than the weighted average, reflecting the disproportionate impact of errors on minority classes.

Attempts to improve accuracy included experimenting with alternative feature representations (TF-IDF vs. CountVectorizer), testing multiple classifiers (Naive Bayes vs. SVM), and enhancing preprocessing with stopword removal, lemmatisation, and explicit negation handling [3]. These steps improved results compared with a simple bag-of-words baseline. Future work could address class imbalance more directly through oversampling, undersampling, or model-based class weighting to improve the classification of neutral reviews.

All experiments are fully reproducible. The Python source code and dataset have been submitted with the report, enabling independent verification and regeneration of the reported results.

4 Conclusion

This project investigated the application of Natural Language Processing (NLP) and machine learning techniques for sentiment analysis of Amazon product reviews, aiming to classify texts into positive, neutral, and negative categories. The study combined two feature extraction methods (TF-IDF and CountVectorizer) with two classifiers (Naive Bayes and Support Vector Machine), following extensive preprocessing that included tokenisation, stopword removal, lemmatisation, and negation handling.

The research questions posed at the outset were addressed as follows. First, the experiments demonstrated that TF-IDF combined with SVM achieved the best performance, with an accuracy of 84% and a macro-F1 score of 0.60 (RQ1). Second, the results showed that SVM produced stable performance across both feature representations, whereas Naive Bayes was highly sensitive, performing poorly with TF-IDF but much better with CountVectorizer (RQ2). Third, the analysis revealed that the neutral class was consistently misclassified across all models due to its under-representation (7.5% of the dataset), confirming that class imbalance significantly affects model performance (RQ3).

In summary, the findings confirm that SVM with TF-IDF is a strong baseline for sentiment classification of product reviews, particularly for positive and negative classes. However, the results also underscore the importance of addressing class imbalance. Future work should explore balancing strategies such as oversampling, undersampling, or classifier weighting, as well as more advanced models such as neural networks, to improve robustness and accuracy for minority classes.

References

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GenAI Acknowledged: This report acknowledges the use of AI tools to support research, grammar checking, and syntax refinement.